

Unit 1: Introduction to HTML

What is HTML?

HTML stands for **HyperText Markup Language**. It is the standard markup language used to create and structure web pages.

HTML describes the structure of a webpage using elements and tags. It tells the browser how content such as text, images, tables, and forms should be displayed.

Example

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Page</title>
</head>
<body>
  <h1>Welcome to HTML</h1>
  <p>This is my first webpage.</p>
</body>
</html>
```

Output:

Welcome to HTML

This is my first webpage.

History of HTML

Year	Version	Description
1991	HTML 1.0	First version developed by Tim Berners-Lee
1995	HTML 2.0	Standard version with basic features
1997	HTML 3.2	Added tables and scripting support
1999	HTML 4.01	Improved styling and multimedia support
2014	HTML5	Modern version with audio, video, canvas, and semantic tags

Father of HTML

Tim Berners-Lee is known as the Father of HTML and the inventor of the World Wide Web.

Structure of an HTML Document

Every HTML page follows a basic structure.

Syntax

```
<!DOCTYPE html>
<html>

  <head>
    <title>Page Title</title>
  </head>

  <body>

    <h1>Main Heading</h1>
    <p>Paragraph Content</p>

  </body>

</html>
```

Explanation

Tag	Purpose
<!DOCTYPE html>	Declares HTML5 document
<html>	Root element
<head>	Contains metadata
<title>	Page title shown in browser tab
<body>	Visible content of webpage

HTML Editors

HTML code can be written using:

1. Notepad
2. Notepad++
3. Visual Studio Code (VS Code)
4. Sublime Text

5. Brackets

Recommended Editor

Visual Studio Code (VS Code)

Features:

- Auto completion
 - Extensions support
 - Live Server
 - Syntax highlighting
-

Creating Your First Web Page

Step 1

Open VS Code.

Step 2

Create a file named:

index.html

Step 3

Type the following code:

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Webpage</title>
</head>
<body>

  <h1>Hello World!</h1>
  <p>Welcome to Full Stack Development.</p>

</body>
</html>
```

Step 4

Save and open in browser.

Output

Hello World!

Welcome to Full Stack Development.

HTML Tags and Elements

HTML Tag

Tags are keywords enclosed within angle brackets.

Examples:

```
<h1>  
<p>  
<body>
```

HTML Element

An HTML element consists of:

```
<tagname> Content </tagname>
```

Example:

```
<p>This is a paragraph</p>
```

Here:

- Opening Tag →
- Content → This is a paragraph
- Closing Tag →

Common HTML Tags

Tag	Description
<h1>	Heading
<p>	Paragraph
	Image
<a>	Hyperlink
<table>	Table
<form>	Form

Attributes

Attributes provide additional information about HTML elements.

Syntax

```
<tagname attribute="value">
```

Example

```
<a href="https://google.com">
  Visit Google
</a>
```

Here:

- href → Attribute
- <https://google.com> → Value

More Examples

```

```

```
<input type="text">
```

```
<p style="color:red;">
  Hello
</p>
```

Common Attributes

Attribute	Purpose
href	Specifies link
src	Specifies image source
alt	Alternative text
style	Inline CSS
id	Unique identifier
class	Group elements

Practical Exercise

Create a Simple Web Page

Program

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Profile</title>
</head>
<body>

  <h1>Harikrishnan</h1>

  <p>
    B.Tech Information Technology Student
  </p>

  <p>
    Welcome to my first HTML webpage.
  </p>

</body>
</html>
```

Output

Harikrishnan

B.Tech Information Technology Student

Welcome to my first HTML webpage.